Open-World Novelty on Transfer Datasets: End-to-End Fine-Tuning, Discovery, and Residual Training

SuperSig experiments 70–72 (with context from 49/51/62/69)

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Abstract

We evaluate six contrastive/regularized pretraining objectives — three unsupervised (SimCLR, SIGReg-SSL/LeJEPA, unsupervised NPLM) and three supervised (SupCon, SupCon+SIGReg, supervised NPLM) — as *end-to-end fine-tuning* losses for ViT-B/16 trunks on four transfer datasets (Stanford Cars, Flowers-102, DTD, Galaxy10 DECals), each from three pretrained bases (DINO, LeJEPA, VISReg): a $4 \times 3 \times 6$ grid. Every space is scored with an open-world novelty battery (held-out–class linear probe, nearest-centroid accuracy, distance-based novelty AUCs, and dataset-level anomaly-detection power), before and after an iterated anchor-*discovery* phase, and finally under *residual* fine-tuning. Three headline results: (i) discovery, which is probe-negative on frozen trunks, becomes probe-positive in 50/72 grid cells once the trunk is fine-tuned end-to-end — and the SupCon-ft+discovery pipeline is positive in 12/12 dataset×base cells; (ii) the champion pre-discovery objective is regime-determined and base-invariant (SupCon+SIGReg on fine-grained data, SimCLR on textures, SupCon on few-class coarse data, supervised NPLM for calibration at few classes); (iii) residual fine-tuning beats the discovery pipeline in all 12 cells, with the NPLM residual concat uniquely raising the probe *while retaining* calibrated per-event power — the first cars space to break the probe–calibration dissociation; (iv) a final discovery pass on the residual winners is *purity-gated*: it stacks only where the residual space separates the novelty, yielding the program's final records — aircraft 0.863, cars 0.855, flowers 0.906, DTD 0.862, Galaxy10 0.975 — every one a residual pipeline.

1 Setup

Datasets and open-world protocol. Each dataset's final classes are held out entirely: their images and labels are removed from every fine-tuning corpus, so held-out classes are genuinely novel at training time.

dataset	classes	held out	train pool (seen)	character
Stanford Cars	196	186–195	7 712	fine-grained (car models)
Flowers-102	102	92–101	1 840	fine-grained, few-shot (∼18/class)
DTD	47	37–46	2 960	texture
Galaxy10 DECals	10	{9}	1 585	coarse morphology, 10%/90% split

Table 1: Datasets. Bases: DINO, LeJEPA and VISReg ViT-B/16 trunks (768-D CLS features).

Fine-tuning recipe (exp-49/62 convention). Model = pretrained trunk + projection head $768 \to 256 \to 100$; everything trainable. Adam with cosine schedule, trunk lr 10^{-5}, head lr 10^{-3}, AMP fp16, 20 epochs, batch 32 with two augmented views (RandomResizedCrop, flip, jitter at 224 px). Per-arm seeds. All objectives operate on the raw 100-D head output $\mathbf{z}$; only softmax-contrastive losses ℓ_2-normalize internally.

2 Loss functions

Let $\mathbf{z}_i = f_\theta(\mathbf{x}_i) \in \mathbb{R}^{100}$, with two views per image stacked into a $2N$ batch; $\hat{\mathbf{z}} = \mathbf{z}/\|\mathbf{z}\|$.

2.1 Core primitives

SIGReg (sliced Gaussianity). The repo's member of the VISReg/LeJEPA family: a sliced 1-D Wasserstein-2 distance to $\mathcal{N}(0, I)$,

$$\mathcal{L}_{\mathrm{SIG}}(\mathbf{z}_{1:B}) \;=\; \frac{1}{S}\sum_{s=1}^{S} \frac{1}{B}\sum_{b=1}^{B}\big(\mathrm{sort}_b(\langle \mathbf{z}, u_s\rangle) - \Phi^{-1}(q_b)\big)^2, \qquad u_s \sim \mathrm{Unif}(\mathbb{S}^{d-1}), \tag{1}$$

with $S{=}64$ slices and $\Phi^{-1}(q_b)$ the standard-normal quantiles. A *classwise* variant applies the same functional per class to $\mathbf{z} - \boldsymbol{\mu}_c$, pulling class c toward $\mathcal{N}(\boldsymbol{\mu}_c, I)$ (used in the CIFAR program; not part of the transfer grid below).

Softmax contrastive (NT-Xent / SupCon). For positives $P(i)$ and temperature τ,

$$\mathcal{L}_{\mathrm{con}} = \sum_i \frac{-1}{|P(i)|}\sum_{p\in P(i)} \log \frac{\exp(\langle \hat{\mathbf{z}}_i, \hat{\mathbf{z}}_p\rangle/\tau)}{\sum_{a\neq i}\exp(\langle \hat{\mathbf{z}}_i, \hat{\mathbf{z}}_a\rangle/\tau)}. \tag{2}$$

With instance positives (the other view) and $\tau{=}0.5$ this is SimCLR/NT-Xent; with class positives and $\tau{=}0.1$ it is SupCon.

2.2 The hybrid contrastive cube (NPLM losses)

`HybridContrastiveLoss` factorizes a contrastive objective into *positives* × *critic* × *estimator* × *marginal*, $\mathcal{L} = \mathcal{L}_{\mathrm{int}}(g) + \lambda\,\mathcal{L}_{\mathrm{marg}}(\mathbf{z})$:

- **critic** $g(\mathbf{z}, \mathbf{z}')$: cosine $\langle \hat{\mathbf{z}}, \hat{\mathbf{z}}'\rangle/\tau$; bilinear $\langle \mathbf{z}, \mathbf{z}'\rangle/\tau$; distance $-\frac{1}{2\tau}\|\mathbf{z} - \mathbf{z}'\|^2$.
- **estimator**: *softmax* (self-normalized per row, as above — recovers pointwise mutual information only up to a per-row constant), or *NPLM*: the un-normalized maximum-likelihood objective

  $$\mathcal{L}_{\mathrm{NPLM}} = \mathbb{E}_{\mathrm{ref}}[e^{\,g} - 1] - \mathbb{E}_{\mathrm{pos}}[g]\,, \tag{3}$$

  whose minimizer is the *absolute* log density ratio $g^* = \log \frac{p(\mathbf{x},\mathbf{x}')}{p(\mathbf{x})p(\mathbf{x}')}$ with the calibration property $\mathbb{E}_{\mathrm{ref}}[e^{g^*}] = 1$ (verified empirically, $r > 0.9$ against analytic PMI). The reference set excludes positives and the diagonal; the exponent is clamped at 30 (row-max subtraction would destroy calibration). At initialization the interaction term is $\mathcal{O}(10^{10})$, so λ is inert early and only shapes the endgame; $\tau = 1$ is a sharp optimum for the bilinear critic.
- **marginal**: none, global SIGReg, or classwise SIGReg.

Calibrated NPLM spaces carry *absolute* distance information (a ready-to-use novelty metric), where softmax spaces are the better linear probe inputs — the program's central probe–calibration dissociation.

2.3 The six fine-tuning arms (exp 70)

arm	objective	labels	knobs
simclr-ft	NT-Xent	no	τ=0.5
sigreg-ssl-ft	$\|\mathbf{z}_1 - \mathbf{z}_2\|^2$ (MSE inv.) + SIGReg	no	LeJEPA objective, λ=1
nplm-bil-ft	NPLM, instance/bilinear + SIGReg	no	τ=1, λ=1
supcon-ft	SupCon	yes	τ=0.1
ss-ft	SupCon + SIGReg	yes	τ=0.1, λ=5
nplm-sup-ft	NPLM, supervised/distance + SIGReg	yes	τ=1, λ=1

Table 2: Pretraining arms: {SimCLR, SIGReg, NPLM} families × {unsupervised, supervised}.

2.4 Residual objectives (exp 71)

Given a trained supervised *parent* with frozen seen-class centroids $\boldsymbol{\mu}_y$ in head space, a deepcopy is fine-tuned end-to-end on the residual $\mathbf{r} = \mathbf{z} - \boldsymbol{\mu}_y$:

$$\textbf{res:}\quad \mathcal{L} = \mathrm{NTXent}_{\tau=0.5}(\hat{\mathbf{r}}) + 5\,\mathcal{L}_{\mathrm{SIG}}(\mathbf{r}) \quad \text{(exp-36/49 champion residual objective)}, \tag{4}$$

$$\textbf{res-nplm:}\quad \mathcal{L} = \mathcal{L}_{\mathrm{NPLM}}^{\mathrm{inst/bilinear}}(\mathbf{r}) + \mathcal{L}_{\mathrm{SIG}}(\mathbf{r}) \quad \text{(exp-59 NPLM residual corner)}. \tag{5}$$

Constructions per cell: supcon-ft→res, ss-ft→res, supcon-ft→res-nplm; each evaluated as the residual head space alone and as the 200-D concat $[\,\mathrm{parent};\mathrm{residual}\,]$.

3 Training strategies and evaluation

End-to-end fine-tuning (exp 70). Each arm fine-tunes trunk+head on the *seen-class* two-view corpus (held-out classes excluded — images and labels), unlike earlier closed-set fine-tunes, so all novelty numbers are strictly open-world.

Discovery (iterated anchor discovery). With the fine-tuned trunk frozen as a feature bank and the head as the discovery backbone: (1) embed the train pool and pool events beyond the 0.95 distance quantile to the current anchors; (2) cluster the pool with BIC-selected k-means, creating new anchors (merged at distance < 3); (3) pseudo-label and fine-tune the head with the proto/repulse hybrid (prototype cross-entropy on $-\frac{1}{2}d^2$ to anchors + repulsion + SIGReg), balanced batches, 2 rounds × 5 epochs. Reported both "naturally" (train pool as-is) and on an injected grid (signal fraction f swept) for post-discovery power.

Metric battery. *probe*: AUC of a linear probe trained to separate held-out from seen test images (3 seeds); *acc*: seen-class nearest-centroid accuracy; *supAUC*: seen-class supervised separability; *eucl*/*mahaT*: novelty AUC of min Euclidean/tied-covariance Mahalanobis distance to seen centroids; *perevt*: per-event detection power at α=0.05 from the distance score; SparKer/Maha/MMD: dataset-level two-sample test power at injected fraction f (toy size n_d=1000–2000; SparKer uses an annealed median-heuristic kernel scale).

arm (DINO base)	cars probe	cars mahaT	flowers probe	flowers mahaT	dtd probe	dtd mahaT	galaxy10 probe	galaxy10 mahaT
simclr-ft	0.624	0.476	0.743	0.673	**0.808**	0.506	0.868	0.730
sigreg-ssl-ft	0.584	0.496	0.732	0.706	0.803	0.577	0.914	0.764
nplm-bil-ft	0.591	0.496	0.664	0.646	0.682	0.479	0.839	0.712
supcon-ft	**0.733**	**0.564**	0.708	0.873	0.799	0.646	**0.938**	0.786
ss-ft	0.719	0.547	**0.795**	**0.883**	0.782	**0.742**	0.872	0.691
nplm-sup-ft	0.549	0.547	0.625	0.668	0.586	0.615	0.878	**0.824**

Table 3: Pre-discovery battery, DINO base (LeJEPA/VISReg bases reproduce the same ordering). The pre-probe champion is regime-determined: fine-grained → ss-ft, texture → simclr-ft, few-class coarse → supcon-ft; supervised NPLM takes the calibration crown exactly at few classes (galaxy10: best eucl/mahaT, per-event 0.35 vs. supcon's 0.02, round-1 pool purity 0.46–0.50 on all three bases).

4 Results

4.1 Pre-discovery: the six arms across the grid

4.2 Discovery: pre → post probe across all bases

arm	cars DINO	cars LeJEPA	cars VISReg	flowers DINO	flowers LeJEPA	flowers VISReg
supcon-ft	.733→**.767**	.699→.726	.707→.759	.708→**.814**	.678→.752	.714→.810
ss-ft	.719→.747	.709→.699	**.741**→.712	**.795**→.788	.792→.613	.754→.743
simclr-ft	.624→.684	.618→.614	.610→.657	.743→.754	.725→.740	.746→.728
sigreg-ssl-ft	.584→.690	.563→.617	.574→.634	.732→.777	.684→.719	.738→.734
nplm-bil-ft	.591→.704	.611→.658	.576→.624	.664→.808	.768→.723	.737→.737
nplm-sup-ft	.549→.600	.582→.658	.579→.549	.625→.624	.734→.706	.631→.576

arm	dtd DINO	dtd LeJEPA	dtd VISReg	galaxy10 DINO	galaxy10 LeJEPA	galaxy10 VISReg
supcon-ft	.799→**.811**	.755→.790	.750→.791	**.938**→.939	.919→.931	**.939**→.944
ss-ft	.782→.785	.787→.784	.784→.785	.872→.926	.803→.871	.865→**.945**
simclr-ft	**.808**→.802	**.817**→.821	**.854**→.848	.868→**.942**	.902→**.946**	.933→.922
sigreg-ssl-ft	.803→.802	.809→.826	.808→.790	.914→.931	**.927**→.940	.931→.940
nplm-bil-ft	.682→.786	.746→.745	.778→.811	.839→.928	.893→.905	.802→.904
nplm-sup-ft	.586→.781	.581→.765	.569→.707	.878→.912	.899→.897	.883→.933

Table 4: Held-out probe, pre → post discovery, full $4 \times 3 \times 6$ grid. Discovery is positive in 50/72 cells; **supcon-ft+discovery is positive in 12/12 dataset×base cells** (the universally safe pipeline). On *frozen* trunks (exp 69, cars) discovery was probe-negative for every arm — the substrate, not the dataset, decides. Notable: the nplm-sup-ft "rescue" on DTD (+0.14 to +0.20 on every base); the one catastrophic case is ss-ft on flowers/LeJEPA (−0.18).

4.3 Residual fine-tuning (exp 71)

The flagship space. Cars/VISReg supcon→res-nplm concat: probe 0.855, nearest-centroid acc 0.608, eucl 0.747, mahaT 0.678, per-event 0.263 — the best cars value in *every column simultaneously*, and the first cars space to break the probe–calibration dissociation. In general the NPLM residual concat raises the probe while *keeping* the parent's calibration, whereas plain-res concats often buy the probe with calibration (galaxy10/LeJePA: probe 0.975 with per-event 0.001).

dataset	base	best construction	probe	perevt	parent	prior champ
aircraft	DINO	supcon→res-nplm concat	0.816	0.225	0.761	0.812†
aircraft	LeJEPA	supcon→res-nplm concat	0.848	0.198	0.799	0.812†
aircraft	VISReg	supcon→res-nplm concat	**0.863**	**0.363**	0.809	0.812†
cars	DINO	supcon→res-nplm concat	0.821	0.148	0.733	0.767
cars	LeJEPA	supcon→res-nplm concat	0.779	0.082	0.699	0.726
cars	VISReg	supcon→res-nplm concat	**0.855**	**0.263**	0.707	0.759
flowers	DINO	supcon→res-nplm concat	**0.885**	0.334	0.708	0.814
flowers	LeJEPA	ss→res concat	0.813	0.258	0.792	0.752
flowers	VISReg	supcon→res-nplm concat	0.858	0.455	0.714	0.810
dtd	DINO	supcon→res concat	0.845	0.045	0.799	0.811
dtd	LeJEPA	ss→res concat	0.831	0.158	0.787	0.826
dtd	VISReg	supcon→res concat	0.847	0.077	0.750	0.854*
galaxy10	DINO	supcon→res concat	0.955	0.021	0.938	0.942
galaxy10	LeJEPA	supcon→res concat	**0.975**	0.001	0.919	0.946
galaxy10	VISReg	supcon→res residual	0.965	0.243	0.939	0.945

Table 5: Best residual construction per cell (probe; per-event power at α=0.05), against its parent and the best prior (exp-70) pipeline. *DTD's prior champion is the *pre*-discovery simclr-ft space — the only exp-70 record residual training does not beat. †Aircraft's prior best carried a trunk-contamination caveat (the exp-62 trunks saw held-out images and labels); the aircraft rows here use new strictly open-world parents (exp-70 protocol, holdouts 90–99 excluded), so 0.863 is both a record and a stricter number. Residual fine-tuning beats the discovery pipeline in **12/12 cells** (15/15 including aircraft).

4.4 Discovery on the residual winners (exp 72)

The feature-space discovery loop was finally run on each cell's exp-71 champion (for concat spaces the discovery backbone is a two-head wrapper — parent head on the parent-trunk bank, residual head on the child-trunk bank — and both heads fine-tune).

Aircraft sharpens the purity gate. Discovery on the three aircraft res-nplm-concat winners is probe-negative (−0.012 to −0.037) despite *moderate* pool purity (0.32–0.53), while buying geometry and per-event power (VISReg: per-event 0.30–0.50 at every fraction, mahaT +0.017): purity is necessary but *not sufficient* — on fine-grained data the res-nplm concat's probe directions are exactly what the discovery fine-tune erodes.

Final dataset probe records (all residual pipelines). aircraft **0.863** (VISReg supcon→res-nplm concat, exp 71, strict open-world parents); cars **0.855** (VISReg supcon→res-nplm concat, exp 71); flowers **0.906** (DINO supcon→res-nplm concat + discovery, exp 72); DTD **0.862** (VISReg supcon→res concat + discovery, exp 72); Galaxy10 **0.975** (LeJEPA supcon→res concat, exp 71).

5 Verdicts

1. **Substrate decides whether discovery helps.** On frozen trunks, fine-grained novelty is not geometrically outlying and discovery erodes the space; on task-fine-tuned trunks the same impure-pool fine-tune acts as refinement (50/72 cells positive; pool purity can stay low). supcon-ft+discovery is the only universally safe pipeline (12/12).
2. **Champion objectives are regime-determined and base-invariant.** Fine-grained many-class → SupCon+SIGReg; texture → SimCLR (labels add nothing: DTD record 0.854 is

cell	space	probe pre → post	mahaT pre → post	purity r_1
flowers/DINO	supcon→res-nplm concat	0.885 → **0.906**	0.843 → 0.853	0.609
flowers/VISReg	supcon→res-nplm concat	0.858 → 0.866	0.870 → 0.875	0.643
dtd/VISReg	supcon→res concat	0.847 → **0.862**	0.758 → 0.778	0.292
dtd/LeJEPA	ss→res concat	0.831 → 0.838	0.791 → 0.796	0.588
cars/LeJEPA	supcon→res-nplm concat	0.779 → 0.787	0.514 → 0.534	0.104
cars/DINO, VISReg	res-nplm concats	flat (−0.001/−0.004)	∼flat	0.16/0.26
flowers/LeJEPA	ss→res concat	flat	flat	0.439
dtd/DINO	supcon→res concat	0.845 → 0.838	0.604 → 0.646	0.135
galaxy10 (3 bases)	res concats / residual	flat to −0.013	mixed	0.00–0.34

Table 6: Exp 72: discovery applied to the best residual space of each cell. Gains are *purity-gated*: discovery stacks only where the residual space separates the novelty (flowers, DTD), and saturates elsewhere (cars, galaxy10 — their exp-71 records stand). New records: flowers 0.906 (first flowers space past 0.90) and DTD 0.862, retiring simclr-ft's 0.854 — the last pre-discovery record. Calibration highlight: galaxy10/VISReg residual + discovery holds per-event power 0.32–0.45 at every injected fraction including f=0.003, the best low-f per-event on record.

unsupervised); few-class coarse → SupCon for the probe and supervised distance-NPLM for calibration — the CIFAR class-count rule, verbatim, on transfer data.

3. **Residual training > discovery everywhere, and discovery on top of residuals is purity-gated.** Residual records: cars 0.855 (+0.088 over discovery), flowers 0.885, galaxy10 0.975; a further discovery pass stacks only where the residual space separates the novelty, adding flowers → 0.906 and DTD → 0.862 (exp 72). The winning residual objective splits by regime (res-nplm on fine-grained, plain res on texture/coarse); supcon-ft is the universal parent, and ss-ft parents fail on fine-grained data because their SIGReg marginal has already absorbed the residual directions.

4. **The probe–calibration dissociation is real but not absolute**: NPLM spaces are the calibrated, distance-ready ones throughout, softmax spaces the probe-ready ones — and the res-nplm concat is the one construction found so far that delivers both at once.

5. **End-to-end fine-tuning erases the frozen-base ranking** (VISReg, weakest frozen, holds several pre records), extending the exp-43/49 inversion to the open-world battery.

Artifacts: scripts `experiments/70_cars_ft_suite.py`, `experiments/71_residual_ft_grid.py`, `experiments/72_residual_discovery.py`; per-cell logs and npz archives in `logs/exp70–exp72`; full tables in `docs/AIRCRAFT_MASTER_TABLE.md`; checkpoints `checkpoints/{ds}_ft_{base}_{arm}_seen.pt`.